
POPCAST: A LEAKAGE-AWARE BENCHMARK FOR AGENTIC FORECASTING ON POP-CULTURE PREDICTION MARKETS

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ABSTRACT

We introduce POPCAST, a benchmark for evaluating language agents on calibrated forecasting under historical information constraints. Unlike static question-answering or tool-use benchmarks, POPCAST asks agents to produce probability forecasts for pop-culture prediction-market events using only information that would have been available at a fixed decision timestamp. The benchmark contains 10 binary tasks spanning film, television, music, awards, celebrity activity, streaming, and viral media. Each task is represented as a frozen JSON market snapshot with explicit resolution criteria, price history, context, volatility and liquidity proxies, a hidden outcome, and post-decision notes withheld from the agent. Agents are scored using Brier score, log loss, and task-success rate. We validate POPCAST by evaluating five systems across 130 total runs: a random prior, a single-prompt frontier LLM, a tool-enabled frontier LLM, a market-only ablation, and the full KALIBRATE agent. The benchmark separates these systems clearly: mean Brier scores range from 0.076 to 0.250, and hard-task performance exposes large failures in retrieval-heavy systems. The most important finding is that market-only forecasting outperforms full retrieval in aggregate, showing that POPCAST captures a real risk in agentic forecasting: web evidence can amplify rumor, hype, and temporal leakage. We close by identifying the benchmark’s main limitation and future direction: fully frozen evidence snapshots for retrieved documents, not only frozen market tasks.

1 INTRODUCTION

Forecasting is a natural setting for evaluating language agents because it requires more than retrieving a fact. A useful forecaster must decide what is knowable at a point in time, reason under uncertainty, and express a calibrated probability. This is especially difficult in pop-culture prediction markets, where events are influenced by announcements, rumors, social media momentum, award campaigns, box-office tracking, streaming performance, and celebrity reporting. Information arrives quickly, but not all information is reliable.

Existing agent benchmarks often evaluate whether an agent can complete a task, answer a question, browse a website, or use a tool. These are valuable capabilities, but they do not directly measure whether the agent can produce a well-calibrated probability about an unresolved event. Forecasting benchmarks do address probability and calibration, but many are built around general news or politics rather than high-noise entertainment domains. Pop culture adds a distinct challenge: search results and media coverage often contain speculation that is directly relevant to the question but not reliable evidence of the outcome.

POPCAST is designed to isolate this setting. Each task is a historical prediction-market-style event with a fixed decision timestamp. The agent receives a sanitized market state and must output a probability for the YES outcome. The benchmark hides the true outcome and any post-decision evaluation notes until scoring. This setup simulates asking: could the agent have made a reasonable forecast at that historical moment, without looking into the future?

Our contributions are:

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1. A 10-task benchmark for pop-culture forecasting across film, television, music, awards, celebrity activity, streaming, and viral media.
 2. A leakage-aware task format that separates agent-visible market state from hidden outcome and post-decision fields.
 3. An evaluation protocol based on proper scoring rules: Brier score and log loss, plus task-success rate for directional interpretation.
 4. A validation study across five systems and 130 runs showing that POPCAST differentiates methods and reveals a key failure mode: retrieval can hurt when evidence is noisy or temporally contaminated.

2 BENCHMARK DESIGN

2.1 DOMAIN AND SCOPE

POPCAST focuses on binary forecasting tasks inspired by Kalshi-style prediction markets in pop culture. Tasks ask questions such as whether a film will cross a box-office threshold, whether an artist will reach a chart position, whether a public figure will announce an engagement, or whether a show will top a streaming chart. These markets are useful for agent evaluation because they are public, outcome-verifiable, time-sensitive, and often noisy.

The benchmark tests four capabilities:

- **Probabilistic reasoning:** producing a calibrated probability, not just a yes/no guess.
- **Temporal reasoning:** respecting what was knowable at a historical decision timestamp.
- **Evidence synthesis:** integrating market prices, context, and optional external evidence.
- **Uncertainty handling:** avoiding overconfident forecasts when the evidence is weak or noisy.

2.2 TASK FORMAT

Each task is stored as a JSON market snapshot. Agent-visible fields include the market title, domain, difficulty, decision timestamp, resolution timestamp, resolution criteria, price history, volatility proxy, liquidity proxy, and context hint. Hidden fields include `outcome` and `notes_for_evaluation`. The market-state constructor removes hidden fields before the task is shown to an agent, and the evaluation harness reads the hidden outcome only after the forecast is produced. A full schema appears in Appendix D.

2.3 TASKS

POPCAST contains 10 tasks: 2 easy, 4 medium, and 4 hard. They cover film box-office thresholds, casting announcements, television release timing, music chart performance, album releases, awards nominations and winners, celebrity announcements, streaming rank, and viral album-debut outcomes. Difficulty labels were assigned before final evaluation based on evidence ambiguity, time-to-resolution, reliance on noisy information, and susceptibility to sudden information shocks. The final results support this stratification: hard tasks create the largest spread across systems. The full task table appears in Appendix C.

2.4 TEMPORAL CUTOFF

POPCAST enforces temporal validity through task-level sanitization, fixed decision timestamps, and retrieval filtering. The raw task JSON contains the outcome and post-decision notes, but these fields are stripped before any agent receives the task. Tool-using systems drop documents published at or after the decision timestamp, and conservatively drop missing or unparseable dates. This prevents obvious post-decision leakage, though our final results show that frozen evidence snapshots are still needed because search snippets and metadata can be unreliable. More detail appears in Appendix A.

Table 1: Overall validation results on POPCAST. Lower Brier and log loss are better; higher TSR is better.

System	Brier ↓	Log Loss ↓	TSR ↑
Random prior	0.250	0.693	0.400
Single-prompt LLM	0.101 ± 0.002	0.344 ± 0.007	0.900
Tool-enabled frontier	0.112 ± 0.007	0.406 ± 0.041	0.900
Market-only ablation	0.076 ± 0.001	0.299 ± 0.003	1.000
KALIBRATE full	0.173 ± 0.006	0.536 ± 0.028	0.800

3 EVALUATION PROTOCOL

3.1 AGENT INTERFACE

An evaluated system receives the sanitized market state and returns a probability $p \in [0, 1]$ that the market resolves YES. Systems may optionally return rationale, confidence, tool traces, and abstention metadata, but only the final probability is scored.

For a live unsettled market, POPCAST would log the forecast and wait until the market resolves. For the current benchmark, all tasks are historical, so every task has a known hidden outcome for scoring while still simulating pre-resolution forecasting.

3.2 METRICS

For binary outcome $y \in \{0, 1\}$ and forecast p , our primary metric is Brier score, $\text{Brier}(p, y) = (p - y)^2$. Lower is better. We also report log loss, which punishes confident wrong forecasts more sharply, and task-success rate (TSR), the fraction of tasks where the forecast is on the correct side of 0.5. TSR is useful for interpretation but is not primary because it ignores calibration: 0.51 and 0.95 are both directionally correct for a YES outcome, but Brier distinguishes them.

3.3 AGGREGATION AND STATISTICAL REPORTING

We report mean metrics across all tasks, plus breakdowns by difficulty and domain. Because POPCAST has only 10 tasks, statistical claims must be modest. For final comparisons, the appropriate test is a paired permutation test over per-task Brier differences between systems, which makes no distributional assumption and respects the paired task structure. Per-domain results often have $n = 1$ or $n = 2$, so they should be reported as exploratory effect sizes rather than significance-tested claims.

4 VALIDATION STUDY

We validate POPCAST using five systems:

- **Random prior:** always predicts 0.5.
- **Single-prompt LLM:** a frontier LLM sees the market state only.
- **Tool-enabled frontier:** a frontier LLM can use Tavily-backed web search up to three times.
- **Market-only ablation:** KALIBRATE with retrieval disabled.
- **KALIBRATE full:** the full structured retrieval and forecasting agent.

The random baseline is deterministic. Each other system is run for three trials per task, producing 130 total runs.

4.1 OVERALL RESULTS

Table 1 shows that POPCAST is not saturated by a random prior: all non-random systems beat 0.250 Brier. It also distinguishes non-random systems. The strongest system is market-only, not the full retrieval agent. This is the benchmark’s most important validation result: it exposes when retrieval hurts rather than merely rewarding agents for using more tools.

4.2 DIFFICULTY AND DOMAIN BREAKDOWN

Difficulty labels matter. Easy and medium tasks are forecastable for most non-random systems, while hard tasks produce the largest separation: KALIBRATE full reaches hard-task Brier 0.377, while market-only remains at 0.118. Domain analysis also shows that market-only does not win uniformly. The full agent is better on film and ties on several domains, while the gap is driven by celebrity and music, where retrieval surfaces rumor-heavy or hype-heavy evidence. This validates POPCAST as a diagnostic benchmark: it does not only rank systems, it reveals where and why they fail. Full breakdown tables appear in Appendix F.

4.3 PER-TASK ANALYSIS

Two tasks illustrate the benchmark’s value. On T6, The Weeknd Grammy nomination task, retrieval helps: the agent finds evidence about his relationship with the Recording Academy and forecasts the correct NO outcome. On T8, the Taylor Swift / Travis Kelce engagement task, retrieval hurts: both tool-using systems become confident in YES even though the outcome is NO. This contrast is exactly what a forecasting benchmark should expose. It is not enough for an agent to retrieve relevant documents; it must judge whether the documents are reliable, temporally valid, and strong enough to move the market prior.

5 CONCLUSION

We introduced POPCAST, a leakage-aware benchmark for calibrated forecasting in pop-culture prediction markets. The validation study shows that the benchmark differentiates systems and reveals meaningful failure modes: the strongest aggregate system is market-only, not the full retrieval agent. This is valuable because POPCAST does not simply reward tool use; it tests whether tools improve calibrated forecasting. In noisy pop-culture markets, evidence can be dangerous when it is speculative, repeated, or temporally contaminated. Future versions should expand the task set, freeze retrieved evidence snapshots, add coverage-risk evaluation, and support sequential market updates.

A LEAKAGE CONTROL DETAILS

Temporal validity is the central benchmark constraint. Every task has a fixed `decision_timestamp`, and all systems receive the same timestamp. For tool-using systems, web-search results are filtered by publication date. Documents published at or after the decision timestamp are dropped before being passed to the agent. Documents with missing or unparseable dates are also dropped conservatively. However, search snippets may summarize future knowledge, pages may have incorrect metadata, and a live search engine may change results over time. The Taylor Swift / Travis Kelce engagement task exposed this risk: tool-using systems appeared to treat engagement information as available before the cutoff. Therefore, the next version of POPCAST should freeze retrieved evidence snapshots as well as task JSON: every URL, snippet, publication date, retrieval time, and cutoff decision should be logged and distributed with the benchmark.

B LIMITATIONS AND FUTURE WORK

POPCAST currently contains 10 tasks, enough for class-scale validation but not for definitive domain-level claims. The context hints make tasks self-contained, but they also encode analyst priors; a deployed market agent would often need to build this context itself. The strongest limitation is evidence freezing. Fixed JSON tasks prevent outcome leakage through the task state, but tool-using systems still depend on live search results. Publication-date filtering reduces leakage but does not eliminate snippet contamination, bad metadata, or changing search rankings. Some frontier models may also have memorized historical outcomes before evaluation. Future work should expand the task set, ship frozen evidence snapshots with every retrieved URL/snippet/date/cutoff decision, add coverage-risk evaluation for abstaining agents, and move toward sequential markets where agents update over time and decide position size.

C TASK LIST

Table 2: Summary of POPCAST tasks.

Task	Domain	Event Type	Difficulty
T1	Film	Box-office threshold	Easy
T2	Film	Casting announcement	Medium
T3	Television	Release timing	Medium
T4	Music	Chart position	Hard
T5	Music	Album release	Easy
T6	Awards	Nomination membership	Hard
T7	Awards	Award winner	Medium
T8	Celebrity	Engagement announcement	Hard
T9	Streaming	Top-10 chart rank	Medium
T10	Viral media	Album debut rank	Hard

D TASK SPECIFICATION TEMPLATE

Each task contains:

```
{
  "task_id": "T1",
  "domain": "...",
  "difficulty": "...",
  "title": "...",
  "decision_timestamp": "...",
  "resolution_timestamp": "...",
  "resolution_criteria": "...",
```

```

270     "outcome": 0 or 1,
271     "price_history": [...],
272     "volatility_proxy": ...,
273     "liquidity_proxy": ...,
274     "context_hint": "...",
275     "notes_for_evaluation": "..."
276 }

```

The fields `outcome` and `notes_for_evaluation` are hidden from agents and used only by the scoring harness.

E SCORING SUMMARY

Given forecast p and outcome y :

$$\text{Brier}(p, y) = (p - y)^2.$$

The final benchmark score is the mean Brier score across tasks. Log loss and TSR are reported as secondary metrics.

F ADDITIONAL RESULTS

Table 3: Brier score by difficulty. Lower is better.

System	Overall	Easy	Medium	Hard
Random prior	0.250	0.250	0.250	0.250
Single-prompt LLM	0.101	0.032	0.116	0.121
Tool-enabled frontier	0.112	0.004	0.006	0.273
Market-only ablation	0.076	0.049	0.046	0.118
KALIBRATE full	0.173	0.018	0.046	0.377

Table 4: Brier score by domain. Lower is better. Domain-level results are exploratory because several domains contain only one task.

System	Awards	Celebrity	Film	Music	Streaming	TV	Viral
Random prior	0.250	0.250	0.250	0.250	0.250	0.250	0.250
Single-prompt LLM	0.017	0.109	0.240	0.091	0.011	0.020	0.176
Tool-enabled frontier	0.007	0.872	0.004	0.046	0.007	0.005	0.124
Market-only ablation	0.023	0.072	0.100	0.071	0.023	0.023	0.250
KALIBRATE full	0.036	0.781	0.070	0.220	0.023	0.023	0.250